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Comparing a general sound quality metric for aircraft noise using a pre-trained neural network model to traditional psychoacoustics

Human Response to UAS and UAM Noise 2

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Abstract When dealing with sound perception, psychoacoustic is often the go-to solution to produce accurate and interpretable results. However, its meaning is very context-dependent, so that a general psychoacoustics-based sound quality metric for different sound types is challenging. This paper investigates a data-driven alternative to obtain sound-quality ratings of various types of aircraft noise using a pre-trained neural network model. As baseline, traditional psychoacoustic approaches are used. The idea of using a pre-trained acoustic model is that it should be able to identify the context where a sound is usually present and thus provide, in general, better predictions. All approaches are trained on diverse listening-test data from various datasets including diverse drones, airplanes and helicopters. Our psychoacoustic baselines use mainly psychoacoustic descriptors from the Sottek Hearing Model (SHM), standardized on ECMA 418-2, as features to a regression model with different degrees of non-linearity, namely: linear models, Kolmogorov-Arnold Networks (KANs), and Tree-based models (XGBoost). Those represent also different levels of interpretability from which an accuracy-interpretability trade-off is identified. Our results indicate that pre-trained acoustic representations can provide a more general solution for aircraft-noise sound-quality prediction, while non-linear regression on psychoacoustic features improves performance relative to linear baselines but may remain constrained by the lack of flexibility/contextual cues. These findings support the use of efficient pre-trained models as a practical route to general sound-quality metrics for aircraft-noise assessment. Additionally, if interpretability is desired, a hybrid approach of identifying annoying sounds with a neural network and then performing a fine-grained analysis with psychoacoustic parameters to evaluate the reason for the perceived annoyance is also a promising approach.

Keywords: UAV Annoyance; Acoustic Pre-trained models; Psychoacoustics; Sound Quality Metrics.

1. INTRODUCTION

Noise impact assessment is increasingly expected to go beyond level-only indicators and incorporate perceptual quality, particularly for emerging air-mobility and Unmanned Aerial Vehicle (UAV) operations. The challenge is not only to analyse average sound levels but to model human annoyance responses to complex spectro-temporal patterns, source dynamics, and contextual variability. Prior aircraft-noise annoyance studies have already shown that a single objective descriptor is often insufficient in mixed-source settings (Green & Torija, 2024). Psychoacoustic modeling remains the standard engineering route because descriptors such as loudness, sharpness, roughness, and fluctuation strength are interpretable and actionable in product/system optimisation workflows. This perspective is also reflected in practical industrial sound quality methodologies (Sottek & Lobato, 2020; Ecma International, 2025). At the same time, broader evidence on aviation noise impact highlights the need for accurate and transferable perceptual predictors for health and quality-of-life motivated assessments (Basner et al., 2017), with UAV-specific studies further highlighting contextual dependence (Christian, 2023). Whilst some studies have sought to better understand the contextual factors of noise perception from Advanced Air Mobility (AAM) operations (Woodcock et al. 2025), it has yet to become common practice for the labelling of said contexts (ambience, vehicle maneouver, time of day) to be included in the training of neural networks, which would presumably increase model accuracy. Other studies have found interaction effects of source-to-environment (Green et al. 2025, Green & Torija 2024), but the psychoacoustic affect of source-to-source interactions between AAMs (temporal overlap) and modelling of masking affects from ambiences remains unknown.In parallel, machine learning for audio modeling is becoming ever more ubiquitous. Tree-based learners and other non-linear regressors can improve over linear models when descriptor interactions matter (Chen & Guestrin, 2016; Liu et al., 2024). More recently, large-scale pre-trained audio representations have become a strong foundation for downstream tasks (Kong et al., 2020; Gong, Chung, & Glass, 2021; Koutini et al., 2022; Chen et al., 2022; Chen et al., 2023; Schmid, Koutini, & Widmer, 2023, 2024). These models can encode time-frequency structures that may be difficult to capture with traditional psychoacoustic metrics alone.

This paper investigates whether pre-trained audio representations can provide a good annoyance predictor even when training for multiple data types in a heterogeneous aircraft-noise setting, while also quantifying how competitive neural networks based on psychoacoustic metrics are in comparison.

The main contribution of this work is a quantitative comparison between psychoacoustic baselines and a fine-tuned pre-trained audio model on a diverse aircraft-noise annoyance corpus. We show that while psychoacoustic regression provides a strong benchmark, the pre-trained approach delivers reliable results faster when sufficient data is available to finetune the pre-trained model.

2. METHODOLOGY

2.1 Dataset

The study uses a combined aircraft-noise annoyance corpus with 635 samples distributed across six listening test datasets as presented in prior work (Green & Torija Martinez, 2024; Ellis et al., 2025). The dataset names and their counts are:

- AJ (KTH): 265
- RN: 120
- MJL: 80
- NG: 71
- D22: 51
- ATM: 48

Annoyance targets are mean ratings per stimulus on a standardized annoyance scale, aggregated over listeners. In this corpus, the annoyance range is 0.39 to 8.90 (mean 4.80, standard deviation 1.57). Fig. 1 summarizes the dataset label distribution.
Results in this paper are evaluated using a 5-fold cross-validation; out-of-fold (OOF) predictions are used to compute all metrics. Fold assignment is performed within each dataset such that each fold contains approximately one-fifth of that dataset. Folds are also stratified by annoyance level, so that no fold contains only low or only high annoyance values.

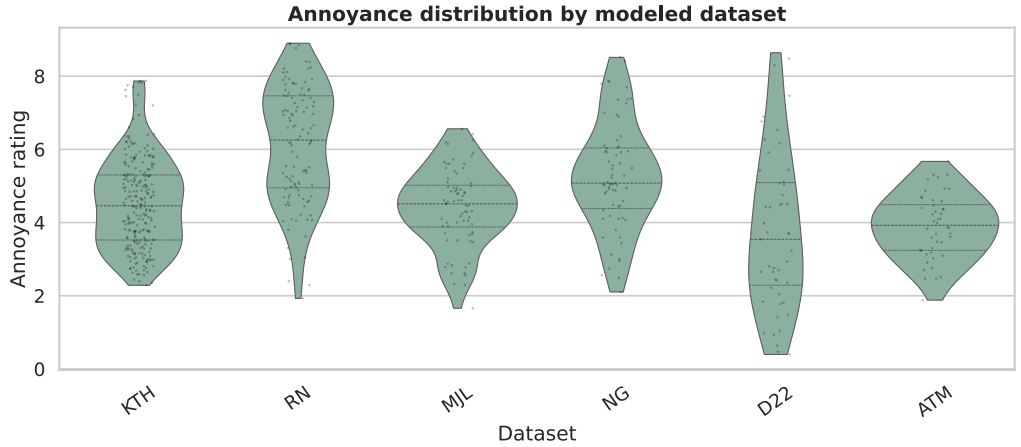


Figure 1: Annoyance distribution by dataset

2.2 Psychoacoustic and descriptor-based baselines

We compute psychoacoustic descriptors using the Sottek Hearing Model (SHM) as specified in ECMA-418-2 (Ecma International, 2025). Additionally, we compute Sharpness using the DIN 45692 standard (DIN, 2009). We evaluated the following regression variants as baselines:

- Linear per-dataset regression: A linear combination of psychoacoustic descriptors fit separately per dataset (the “classical way” of generating sound quality metrics).
- Linear per-dataset SHM-loudness-only regression: A variant using only SHM loudness, as a strong single-feature baseline.
- XGBoost regressor (Chen & Guestrin, 2016): A non-linear regressor using the full psychoacoustic feature set, allowing non-linear interactions between descriptors.
- KAN regressor (Liu et al., 2024): A flexible function approximator over psychoacoustic metrics, aligned with interpretability-oriented variants (Lobato & Kamp, 2025).

This set spans a practical spectrum from high interpretability (single-descriptor and linear mappings) to higher flexibility (tree-based and neural function approximators over descriptors). All model hyperparameters are optimized with a fixed-budget 200 iterations with a Optuna (Akiba, T et al, 2019) search, using the same folds and objective (mean validation MAE).
Linear regression baselines use L1/L2 regularization while for XGBoost we optimized: learning rate, depth, number of estimators, subsampling, and L1/L2 penalties. For KANs we optimized the regularization strength and architecture structure.
To account for possible dataset-level biases in absolute annoyance (e.g., differences in scale usage between listening tests), we allow an additional calibration step in the form of an affine transform $(ax + b)$ for the linear models. This is typical in sound-quality metrics (to translate psychoacoustic to sound quality) and lets us focus on models that can produce accurate rankings while potentially being offset in absolute values. This is a practical solution that allows training from multiple sound quality datasets and creates a model capable of performing relative comparisons. For the non-linear models, especially XGBoost, an end-to-end training with an

affine transformation is challenging. Thus, we approximated it by providing a one-hot-encoding of the dataset as feature for the training (for XGBoost and KANs). Even though the resulting model is not directly usable (i.e. dataset is unknown by inference), they continue to be good baselines for performance.

2.3 Pre-trained audio model

The neural approach follows the latest successes of large-scale pre-trained audio modeling (Kong et al., 2020; Gong, Chung, & Glass, 2021; Koutini et al., 2022; Chen et al., 2022; Chen et al., 2023) and uses a pre-trained EfficientAT representation backbone (a MobileNet-like CNN distilled from transformer models) (Schmid, Koutini, & Widmer, 2023, 2024). The backbone is fine-tuned for annoyance regression with a linear regression head. One additional reason we've chosen this backbone is that it allows a very fast inference on edge devices, creating the possibility of deploying it locally by users, which could be useful for future sound level meter (SLM) devices with low compute to allow for real-time psychoacoustic analyses and prediction Audio samples were resampled to a common sampling rate and converted into fixed-length segments. As input representation, we used a log-scaled time-frequency representation based on mel-frequency bands, followed by normalization.
The model was optimized using a two-stage training procedure. First, the task-specific prediction head was trained while the feature extraction backbone was kept fixed. In a second stage, the complete model was fine-tuned end-to-end. Hyperparameters, including learning-rate schedules and relative learning rates for different model components, were selected using automated validation-based optimization. The regression objective was based on absolute prediction error.
Similarly to the linear regression baselines, the neural model includes a learned per-dataset calibration term (a learnable per-dataset affine transform). The outputs of the model are then "scale-less", in the sense that they don't represent an absolute annoyance scale without an affine calibration (but retain relative order).

3. Results and Discussion

Results are evaluated using 5-fold cross-validation and out-of-fold (OOF) predictions are used to compute all metrics. Fold assignment is performed within each dataset such that each fold contains approximately one-fifth of that dataset. Folds are also stratified by annoyance level (using binning), so that no fold contains only low or only high annoyance values. Metrics are MAE, RMSE and R2.
Figure 2 shows the aggregate results over all datasets. Relative to the strongest baseline (in this case XGBoost), the pre-trained EfficientAT model reduces MAE by ~0.1 (~25%). Relative to linear per-dataset regression, the MAE reduction is ~0.17 (~35%). This is a considerably reduction, showing that annoyance predictions can benefit from pre-trained audio models.

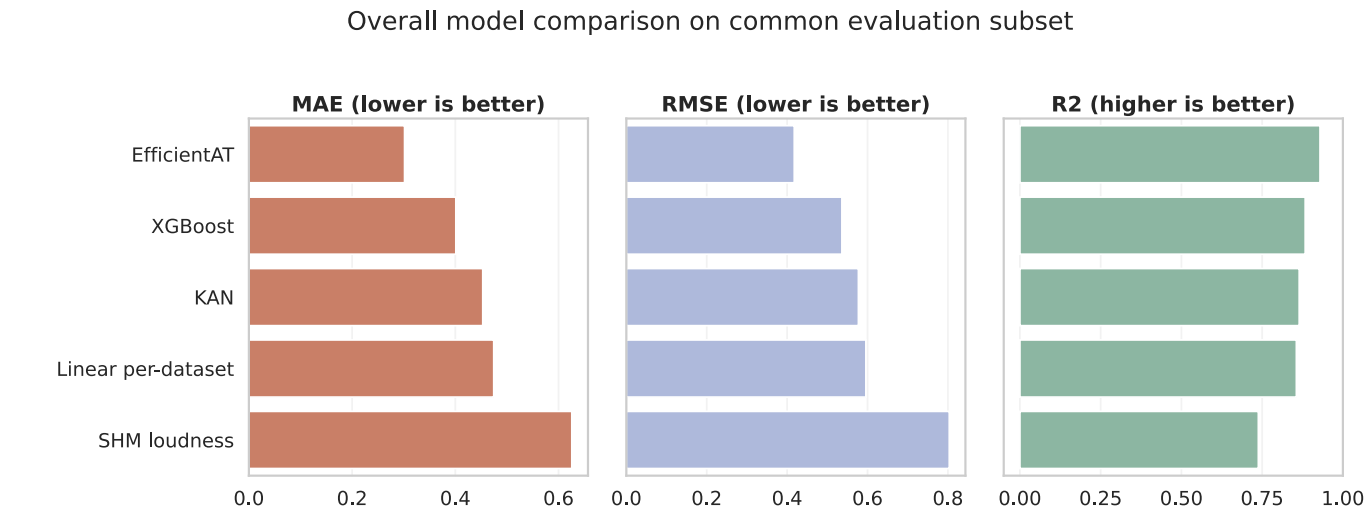


Figure 2: Overall model comparison on the common evaluation subset (MAE, RMSE, R2). Lower is better for MAE/RMSE and higher is better for R2.

When evaluating per dataset, we can see the result in Figure 3, while the approaches with best MAE by dataset are:

- ATM (n=48): KAN (0.43),
- D22 (n=51): Linear per-dataset (0.28),
- KTH (n=265): EfficientAT (0.17),
- MJL (n=80): EfficientAT (0.29),
- NG (n=71): XGBoost (0.39),
- RN (n=120): EfficientAT (0.36).

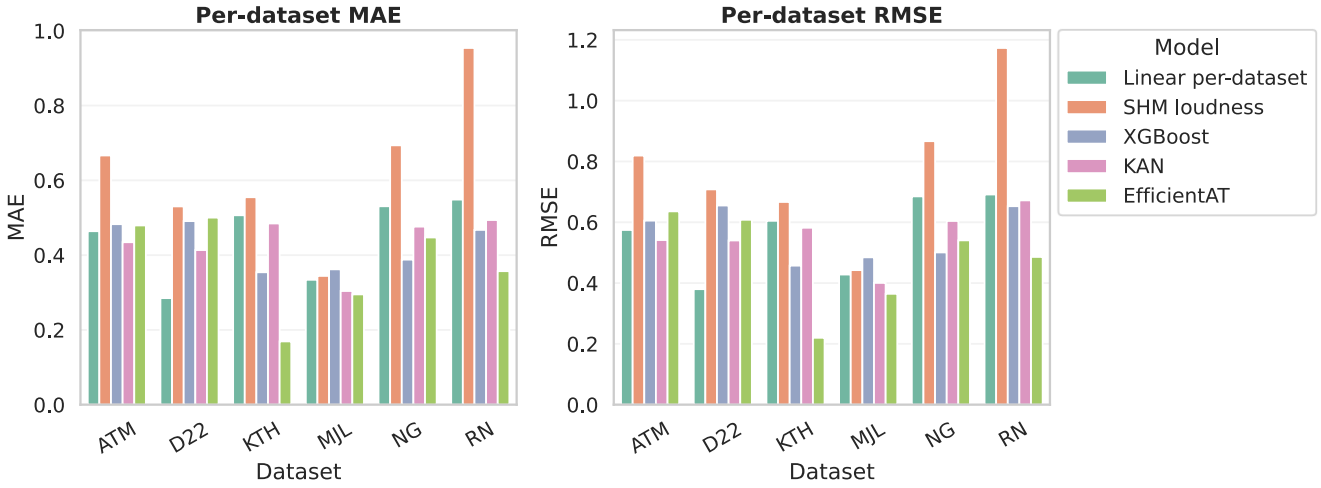


Figure 3: Per-dataset MAE and RMSE bar plots across model families.

Thus, we see that EfficientAT is not uniformly best in every dataset, but it achieves the lowest pooled aggregate error under within-dataset cross-validation. This shows that, depending on the case, classical psychoacoustics still may have the upper hand on the results. A general plot with all data with results from all methods is shown in Figure 4. The fact that there is no “Dominant Approach” was also identified in (Lobato & Kamp, 2025).

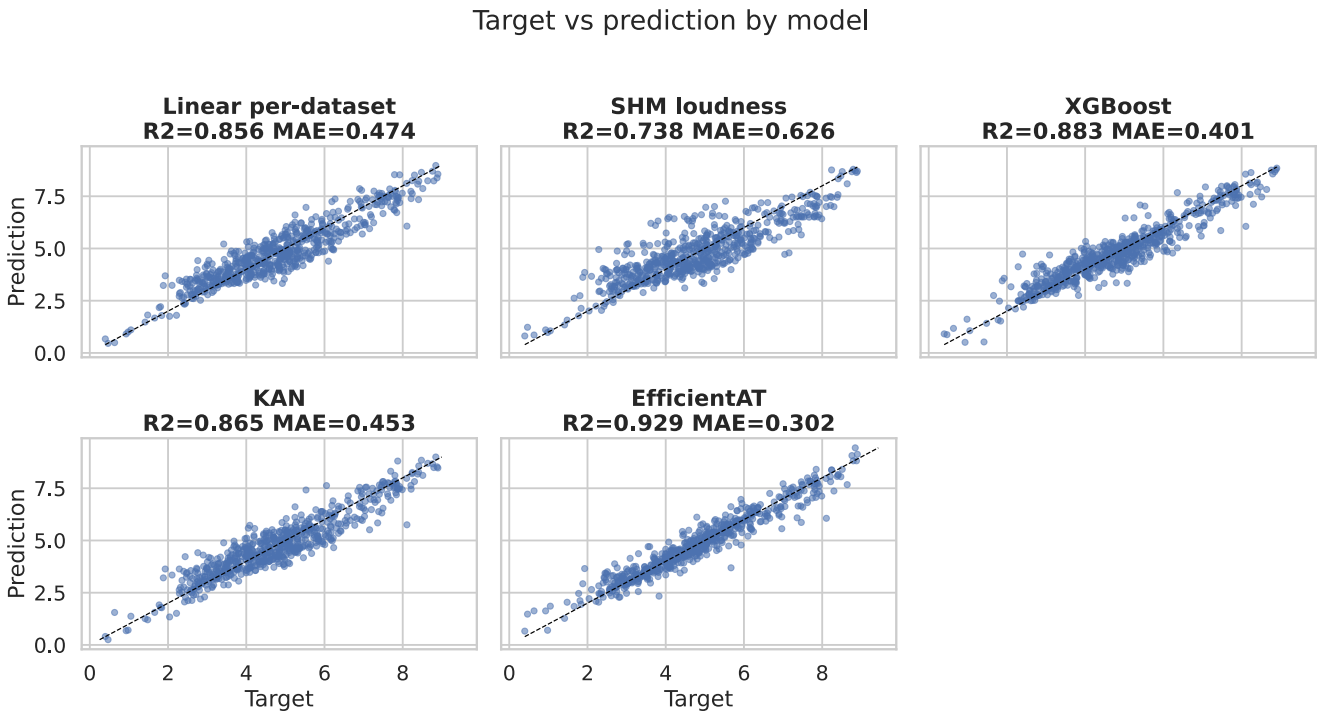


Figure 4: Target-versus-prediction for all compared models on all data

We've seen that pre-trained neural representations can improve the performance of the annoyance prediction. This is likely because they can encode spectro-temporal patterns and higher-order interactions that are not explicitly represented by the selected psychoacoustic set. The advantages of a higher-order interaction can indeed be seen by verifying the better results of non-linear models using the psychoacoustic metrics.

Still, psychoacoustics remains strong and relevant. Linear per-dataset regression is still highly competitive and even shows better results on the ATM and D22 datasets compared to the EfficientAT model (and at D22 an overall better result). We notice that those are the smallest datasets, which may indicate that when data volume is limited, simple linear psychoacoustic models may provide robust performance with the advantage of clear interpretability. This is something that was also observed in (Lobato & Kamp, 2025). This indicates that important sound phenomena relevant to aircraft noise may be explainable by psychoacoustics, allowing their use as a diagnostic tool, possibly preceded by a neural network-based metric.

An interesting result is that the SHM-loudness baseline is noteworthy despite relying only on a single descriptor. While it is worse than multi-feature linear per-dataset and XGBoost in aggregate, its performance remains especially high, suggesting that a large fraction of aircraft annoyance may be explained by Loudness alone. At the same time, additional descriptors and learned representations clearly help reduce absolute error.

The KAN results stay in between the Linear regression and XGBoost variants. This indicates that more complex interactions of psychoacoustic features may be able to improve results while still being interpretable. This is a classical trade-off between interpretability and accuracy of results, in which KANs allow interpretable non-linearity (which, however, may not be as easy to interpret as simple linear coefficients).

Overall, based on our results, we recommend a hybrid workflow:

1. Use pre-trained neural models for high-accuracy prediction.
2. Use psychoacoustic descriptors to interpret and diagnose problematic events. (eventually with KANs for higher order interactions)
3. Use test-specific calibration whenever absolute values are of relevance (possibly based on a small listening test).

4. Conclusion

This work compared linear and non-linear psychoacoustic baselines to a pre-trained neural model for aircraft-noise annoyance prediction under 5-fold cross-validation on a pooled aircraft-noise annoyance corpus. The pre-trained model achieved the strongest aggregate performance (MAE = 0.302, R^2 = 0.929), while psychoacoustic models remained competitive and were better in two datasets while retained interpretability advantages. This is a good indication that psychoacoustic metrics are still relevant for sound annoyance evaluation, especially if interpretability and troubleshooting are needed. Black-box models on the other hand may be used as complementary tools to achieve a quick and accurate result for an annoyance level without any reasoning by psychoacoustic parameters such as for example roughness or tonality.

As general recommendation based on our results, a first quick evaluation with a neural network model for better accuracy followed by psychoacoustic probing for better understanding of the results seems to be a good way to achieve the best results while also understanding its origin.

For the psychoacoustic analysis, we've used default single value predictors. In the future, it may be interesting to see how models behave when giving psychoacoustic results over time to the neural network. This may allow it to perform complex combinations of results or define its own signal-dependent single value, which would leverage neural networks to improve even more the interpretability aspects of psychoacoustics.

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